Models and Training Datasets

PostAssist LK - Smart Registered Mail Intake & Tracking component (J26-IT-372)

1. The three trained models at a glance

The counter photographs an envelope. Three small models turn the photo into a checked address: **(1)** decide the script of each line, **(2)** read handwritten lines, **(3)** split the text into address fields. Every field still has to be confirmed by the officer; the models only save typing and point to fields that need checking.

Model	Task	Algorithm	Trained on	Held-out test result	Shipped file
Script identification	Is a text line Sinhala, Tamil or Latin?	CNN (5 conv blocks + global average pooling), softmax + temperature scaling	45,000 rendered lines, 27 Google Fonts	98.6% accuracy on 6 fonts never seen in training	script_id.onnx (1.9 MB)
Handwriting recogniser (HTR)	Read one handwritten line into characters	CRNN: 7 conv + 2 x BiLSTM + CTC decoding	SinOCR, IAM, Sinhala 454-class characters, fonts	CER Sinhala 18.1% (Tesseract 90.7%); English 9.2% (Tesseract 56.2%)	htr.onnx (18 MB)
Address-field tagger	Label each word NAME, HOUSE, STREET, LOCALITY, CITY, REGION, POSTCODE, COUNTRY, O	Linear-chain CRF (python-crfsuite, L-BFGS)	34,460 addresses: Deepparse real + synthetic	Required-field exact match 99.4% (rule baseline 57.5%), towns never seen	address_crf.crfsuite (0.2 MB)

Code: `ml/script_id/`, `ml/htr/`, `ml/address_parser/` - models are written to `backend/models/`, reports to `ml/reports/`.

2. Source datasets

All sources are public and openly licensed. Every download is recorded in `Datasets/raw/MANIFEST.json` with URL, licence, citation and SHA-256.

Dataset (source)	Size	Columns / structure	Licence	Used for	Why it is a good source
Sri Lanka provinces-districts-cities (M. Madurapperuma, GitHub)	2,155 postal towns	locality_id, name_en/si/ta, sub_name_en/si/ta, postcode, district_en/si/ta, province_en/si/ta, latitude, longitude, coordinate_provenance	MIT	CRF gazetteer features, town-postcode check, synthetic addresses, script-ID text	The only open list of every Sri Lankan postal town in all three languages with its 5-digit postcode
Deepparse worldwide-addresses, lk split (Yassine & Beauchemin, Hugging Face)	19,292 addresses	Address (text), Tags (one tag per word: StreetNumber, StreetName, Municipality, PostalCode, Country...), Language	CC BY 4.0	The real-address part of the CRF corpus	Real Sri Lankan addresses that are already labelled word by word - exactly what a field tagger needs
OpenStreetMap (Overpass API)	797 post offices, 358 streets	Post offices: osm_ref, name, name_en/si/ta, postcode, city, latitude, longitude. Streets: name_en/si/ta, highway	ODbL 1.0	Street vocabulary, facility reference points	Real street names written in Sinhala and Tamil, and real post-office positions
Wikidata (SPARQL)	1,323 names (with seed list)	lang, kind (given / family), name, source	CC0 1.0	Recipient names in synthetic addresses	Real Sri Lankan names; public domain, no personal data linked to an address
Google Fonts (Noto, Abhaya Libre, Yaldevi, Hind Madurai, Caveat, Kalam, Indie Flower...)	27 fonts	TrueType files (Sinhala, Tamil, Latin, handwriting-style)	SIL OFL 1.1	Rendering script-ID lines, synthetic HTR lines, synthetic envelopes	Many typefaces per script, so the model learns the script and not one font; fonts can be held out for a fair test
SinOCR - handwritten subset (IEEE DataPort)	1,135 lines (908 train / 227 test)	data.csv: file_name, text + line images	As on IEEE DataPort (confirm before publishing)	Real Sinhala handwriting: train, validation (10% of train), test	The only public set of real handwritten Sinhala lines; it contains names and addresses from forms
Sinhala handwritten characters, 454 classes (Dataset454)	108,933 images, 80 x 80 px	Folders train / valid / test / <class 1-454> / *.jpg; no label file (270 classes labelled by eye in sinhala_glyphs.py)	To be confirmed by the project owner	Synthetic Sinhala lines: address text spelled with real handwritten letters	Real Sinhala pen strokes from many writers, so synthetic lines look handwritten

Dataset (source)	Size	Columns / structure	Licence	Used for	Why it is a good source
IAM handwriting database - lines (Marti & Bunke 2002; Hugging Face Teklia/IAM-line)	10,373 lines (6,482 / 976 / 2,915)	Parquet: text, image	Free for non -commercial research	Real English handwriting: train, validation, test	The standard benchmark for English line handwriting; results are comparable with published work
Department of Posts Sri Lanka - postage revision, 9 Feb 2026	21 weight bands	JSON: tariff_version, bands (weight from/to, postage), registration fee, advice of delivery, insurance	Public government notice	Price calculation (not model training)	The official current tariff

3. Training datasets built from the sources

Processed dataset	Model	Size and split	Fields / columns
Datasets/processed/address_tagging/*.jsonl	CRF	34,460 addresses: train 24,245 / val 4,878 / test 5,337. Split 70/15/15 **grouped by postal town**. Train: 7,358 real + 16,887 synthetic; ~25% with OCR-style noise	text, labels (one per word), source (deepparse_lk / synthetic), script (si / ta / en / mixed), group (town), noisy
Datasets/processed/script_id/*.npz	Script-ID CNN	45,000 / 4,500 / 4,500 lines, 32 x 256 px. Train 18 fonts, validation 4 fonts, test 6 **held-out fonts** + held-out address text	image (grey line), label (si / ta / en), font, style
HTR cache (%USERPROFILE%\PostAssist\datasets\cache)	HTR	Real: SinOCR 820 / 88 / 227, IAM 6,482 / 976 / 2,915 (test sample 400). Each epoch 20,362 lines	line image (48 px high) + text
Datasets/processed/synthetic_envelopes/annotations.jsonl	Evaluation only	360 envelope photos (120 per script, 60 printed + 60 handwriting-style)	image_id, file, script, style, font, writer_code, base_address_id, split, recipient_text, recipient {name, house_number, street, locality, city, region, postcode, country}

Why synthetic data is added: no public dataset has labelled Sinhala and Tamil envelope addresses or Tamil handwriting. Synthetic addresses are built from real towns, postcodes, streets and names; given names, family names, house numbers and streets are drawn independently, so no real person is linked to a real address.

4. Each model in detail

4.1 Address-field tagger - linear-chain CRF

- **Algorithm:** Conditional Random Field over the word sequence; trained with L-BFGS (python-crfsuite). It picks the best label for every word while considering the labels of its neighbours.
- **Why a CRF:** works well on small data; the features are explainable; it gives a calibrated probability per word (used as field confidence); takes milliseconds on a counter PC with no GPU.
- **Features per word:** the word, prefix/suffix, script (Sinhala/Tamil/Latin), shape, digits, slash/hyphen/comma, capitals, gazetteer membership (town, district, province, known postcode), trilingual vocabulary (honorifics, house markers, street types), line position on the envelope, words +/-2 around it.
- **Parameters:** grid search over 5 (c1, c2) pairs on an 8,000-address subsample; selected **c1 = 0.1 (L1), c2 = 0.01 (L2)**; max 200 iterations; seed 7.
- **Confidence:** field confidence = lowest word probability in the field. Review threshold **0.974** (the smallest threshold at which unflagged validation fields are at least 99.7% correct).
- **Result (test, unseen towns):** token accuracy 99.7%, macro-F1 99.5%, required-field exact match 99.4%, whole-address exact match 98.3%, ECE 0.0024. Real Deepparse rows: 97.6% required-field exact match.

4.2 Script identification - CNN

- **Algorithm:** convolutional neural network: 5 blocks of Conv 3x3 + BatchNorm + ReLU, global average pooling, linear layer to 3 classes; cross-entropy loss. **462,755 parameters**; input 1 x 32 x 256.
- **Why a CNN:** the three scripts differ in shape (Sinhala curls, Tamil loops, Latin strokes), which is what convolutions detect. Small enough for about 7 ms per line on CPU (ONNX Runtime).
- **Parameters:** AdamW, learning rate 0.002, weight decay 0.0001, OneCycle schedule, 14 epochs, batch 128. Three seeds (5, 6, 7); the one with the best *validation* accuracy (seed 6) is shipped.
- **Calibration:** temperature scaling (Guo et al., 2017), T = 0.5468 fitted on validation. Calibration error on test 0.0667 -> 0.0115.
- **Result:** 98.6% test accuracy on unseen fonts; across seeds 97.5% +/- 1.4%.

4.3 Handwriting recogniser - CRNN + CTC

- **Algorithm:** CRNN: 7 convolution layers (see the strokes) -> 2-layer bidirectional LSTM, 256 units, dropout 0.25 (reads left to right and right to left) -> linear layer over 195 characters + blank. Trained with **CTC loss**, decoded greedily. **4.48 M parameters**. Input: line 48 px high, any width; one output frame per 4 px.
- **Why CRNN + CTC:** the standard method for line handwriting; it needs only the line text, not the position of every letter; one network reads Sinhala, English and Tamil, so mixed lines need no switching.
- **Parameters:** AdamW, learning rate 0.0015, weight decay 0.0001, warm-up then cosine decay, gradient clipping 5.0, batch 32, 40 epochs (best epoch 37), mixed precision on an RTX 2050, seed 372.
- **Lines per epoch (20,362):** IAM 6,482 + SinOCR 820 x 4 + 6,000 Sinhala lines built from 454-class characters + 600 Sinhala font + 2,000 English font + 2,000 Tamil font. Synthetic lines are drawn fresh every epoch, with augmentation.
- **Selection:** lowest mean character error rate on validation (real Sinhala, real English, Tamil in a validation font). Test sets read once, after training.
- **Result:** CER Sinhala 18.1%, English 9.2%, Tamil (unseen fonts) 0.4% - about 5x fewer errors than Tesseract on handwriting.

5. How the testing was kept fair

- Three separate splits: **train** (learn), **validation** (choose settings and threshold), **test** (read once, at the end).
- The test data is genuinely new: **unseen towns** (CRF), **unseen fonts** (script-ID), the **published test splits** of SinOCR and IAM (HTR).
- Compared with baselines: a rule-based tagger (CRF) and Tesseract 5.5 (HTR).
- Fixed random seeds, so every split and model can be rebuilt with `bash ml/run_all.sh`.

6. Limitations (state them before you are asked)

- Most Sinhala and Tamil address data is **synthetic**; results on real mail may be lower. Quote the real-address row first.
- **No real Tamil handwriting** was used for training or testing; Tamil was learnt from fonts only.
- The 454-class character labels were assigned by eye; the licences of SinOCR and Dataset454 must be confirmed before publishing.
- Final claims need the consented real-envelope set (`ml/evaluation/ocr_benchmark.py`).